



MODULE NAME:	MODULE CODE:
MATHEMATICAL PRINCIPLES FOR COMPUTER SCIENCE	MAPC5112/p

ASSESSMENT TYPE: ASSIGNMENT 1 (PAPER ONLY)

TOTAL MARK ALLOCATION: 100 MARKS

TOTAL HOURS: 10 HOURS

By submitting this assignment, you acknowledge that you have read and understood all the rules as per the terms in the registration contract, in particular the assignment and assessment rules in The IIE Assessment Strategy and Policy (IIE009), the intellectual integrity and plagiarism rules in the Intellectual Integrity and Property Rights Policy (IIE023), as well as any rules and regulations published in the student portal.

INSTRUCTIONS:

1. ***No material may be copied from original sources, even if referenced correctly, unless it is a direct quote indicated with quotation marks. No more than 10% of the assignment may consist of direct quotes.***
2. ***Your assignment must be submitted through Turnitin.***
3. ***Save a copy of your assignment before submitting it.***
4. ***Assignments must be typed unless otherwise specified.***
5. ***All work must be adequately and correctly referenced.***
6. ***This is an individual assignment.***

Referencing Rubric

Providing evidence based on valid and referenced academic sources is a fundamental educational principle and the cornerstone of high-quality academic work. Part of achieving this quality is referencing in a way that is consistent and congruent with the requirements of the referencing style being used.

Therefore, inconsistent and/or incongruent referencing will result in a penalty of **a maximum of ten percent being deducted from the overall percentage** awarded to your assessment submission.

Please note that **evidence of plagiarism** in the form of copied or unreferenced work, absent reference lists, or exceptionally poor referencing **may result in action being taken in accordance with The IIE's Intellectual Integrity and Property Rights Policy (IIE023)**. Similarly, **evidence of excessive AI usage may result in action being taken in accordance with The IIE's Student Conduct, Discipline and Safety Policy (IIE015)**.

Markers are required to provide feedback to students by **circling/underlining the information in the table below that best describes the student's work and by adding constructive commentary where appropriate**. The examples provided are not exhaustive but illustrate the errors.

Deductions

- Where the student's work contains **five or more errors** aligned to the **minor errors column** below, **deduct 5% from the overall percentage**.
- Where the student's work contains **five or more errors** aligned to the **major errors column** below, **deduct 10% from the overall percentage**.
- Where both minor and major errors** (e.g. two minor and three major, etc.) are present, **deduct 10% only** (and not 5% or 15%) from the overall percentage.

Required: Consistent and congruent referencing	Minor errors Deduct 5% from overall percentage. Example: if the response receives 70%, deduct 5%. The final mark is 65%.	Major errors Deduct 10% from the overall percentage. Example: if the response receives 70%, deduct 10%. The final mark is 60%.
Consistency - The correct referencing style for the discipline – i.e., either Harvard, OR APA (for Psychology), OR Law, OR IEEE (for ICT/Engineering) – has been used consistently for all in-text references and in the bibliography/reference list. Concepts and ideas that are quoted and/or paraphrased are referenced consistently throughout. Position of the in-text reference: an in-text reference is positioned consistently where appropriate for every quote and paraphrase.	Minor inconsistencies: - The referencing style used is generally consistent with what is required, but there are one or two changes/errors in the format of in-text referencing and/or in the bibliography/reference list. - For example, page numbers for direct quotes in-text have been provided for one source, but not in another. Or, two book chapters in the bibliography/reference list have been referenced in two different formats. Or, the publication year has been placed after the author name in one bibliography/reference list entry, and after the source title in another, etc. - Concepts and ideas in quotes and/or paraphrases are typically referenced, but a full in-text reference is missing or incomplete from one or two small sections of the work. - Position of the references: in-text references are only given at the beginning and/or end of every paragraph.	Major inconsistencies: - Poor and wholly inconsistent referencing style used in-text and/or in the bibliography/reference list. - Multiple referencing styles for the same source types have been used. - For example, the format for direct quotes in-text and/or book chapters in the bibliography/reference list and/or year of publication in the bibliography/reference list is different across multiple instances. - Concepts and ideas in quotes and/or paraphrases are haphazardly referenced in-text. - Position of the references: in-text references are only given at the beginning or end of large sections of work.
Feedback on referencing consistency:		
Congruency Each source reflected within in-text references is included accurately in the bibliography/reference list. - All bibliography/reference list entries are in the required order for the referencing style used (e.g. alphabetical, alphabetical under subheadings, numerical). - All direct quotes and paraphrases have been integrated appropriately into the text using introductory phrases, accurate grammar, etc.	Minor incongruities: - There is largely a match between the sources presented in-text and those in the bibliography/reference list, but one or two sources that appear in-text do not appear in the bibliography/reference list, or vice versa. Or key source information is missing from one or two in-text references or bibliography/reference list entries only (e.g. publication year, city of publication, URL date accessed, etc.). - There is a clear and largely accurate ordering of sources in the bibliography/reference list as required by the referencing style used, but with one or two references out of order. - An attempt has been made for source integration into the text using appropriate introductory phrases and grammar, but one or two quotes or paraphrases do not flow as clearly or logically within the sentence structure as they could.	Major incongruities: - No relationship/several incongruities between the in-text referencing and the bibliography/reference list. - For example, multiple sources are included in-text, but not in the bibliography, and/or vice versa. Key source information is missing from multiple in-text references and/or reference list entries. A URL link, rather than the actual reference, is provided in the bibliography. Sources are repeated in the reference list, etc. - Most sources are listed in a haphazard order throughout the bibliography/reference list. - Few to no appropriate introductory phrases or rules of grammar have been applied, and many direct quotes and/or paraphrases feel disconnected from the flow of the text.
Feedback on referencing congruency:		
Overall feedback on referencing, with suggested improvements:		

Question 1		(Marks: 25)
Q.1.1	Given the arithmetic expression:	
$8 + 6 \div 2 \times (3 + 5) - 4$		
Q.1.1.1	Simplify the expression step-by-step following the order of operations (PEMDAS/BODMAS).	(6)
Q.1.1.2	Verify if the final result is correct by evaluating the expression using a different order (incorrect) and explaining any differences.	(9)
Q.1.2	Using the relevant exponential rules, simplify the following expression, leaving your final answer in exponential form:	(10)
$\frac{2^3 \cdot 4^{-1} \cdot 8^2}{16^{-1} \cdot 32^{\left(\frac{1}{5}\right)}}$		
Provide a step-by-step solution to reach the simplified form.		
Note: Marks will not be awarded for using a calculator, such as converting 2^3 to 8. Demonstrating the rules of exponents will earn you full points.		

Question 2		(Marks: 40)
Q.2.1	Solve the following equation and simplify your answer using the properties of equalities. $\frac{4x - 3}{5} + \frac{5x + 9}{4} = \frac{2x}{3} + \frac{15}{2} - \frac{7x}{6}$ Show all your steps and leave your answer as an improper fraction.	(6)
Q.2.2	Consider the absolute value inequality: $ 5 - 4x \leq 15$ Find the solution set for the absolute value inequality and present your answer as one inequality. Show all your steps.	(8)
Q.2.3	If the product of two numbers is equal to 60 and their average is equal to 8, find both numbers using quadratic equations as part of your steps. Show all your steps.	(10)
Q.2.4	Simplify the following expression, ensuring to account for both odd and even roots: $\sqrt[3]{-8a^3 b^6} \times \sqrt[4]{16x^8 y^{12}}$ Show all your steps and explain your reasoning.	(8)
Q.2.5	Solve the following radical equation and provide verification of your solution in a step-by-step format: $3 + \sqrt{4x - 5} = 8.$ Show all your steps.	(8)

Question 3		(Marks: 35)
	Q.3.1 In the context of programming, provide one example of an atomic statement and one example of a molecular statement. For each example, justify why it is considered atomic or molecular based on its structure and meaning. Note: - Do not provide textbook definitions. - Marks will be awarded based on the relevance and clarity of the justification provided for each example.	(10)
	Q.3.2 Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 6\}$, $C = \{1, 2, 3\}$, and $D = \{7, 8, 9\}$. The universe set is $U = \{1, 2, \dots, 10\}$. Find the following:	
	Q.3.2.1 $(B \cup C)^c$	(3)
	Q.3.2.2 $A \setminus B$	(3)
	Q.3.2.3 $(D \cap C^c) \cup (A \cap B)^c$	(9)
	Q.3.3 Use the formulas $T_n = \frac{n(n+1)}{2}$ and $a_n = 2^n$ to find closed formulas for the following sequence: (c_n): 0, 1, 3, 7, 15, 31, ... Instructions: - Identify the pattern in the given sequence. - Determine the closed formula for the sequence (c_n). - Verify the closed formula by calculating the first few terms and comparing them with the given sequence. - Show all your steps and reasoning clearly.	(10)

END OF PAPER